

NLP Project Report – Part 1

September 2025

Abstract

This report presents our initial approach to the problem of predicting attribute values from product text in e-commerce catalogs. We focus on the IndoML Datathon dataset, which contains both product titles and receipt-style descriptions. Our plan is to begin with simple NLP baseline models and then progress to transformer-based and LLM-based models later.

1 Introduction

E-commerce platforms depend on structured product attributes such as category and brand to organize catalogs and power search and recommendations. However, product listings often contain only short and noisy text, and many sellers avoid filling in all attributes due to the manual effort required. This results in incomplete product listings, making it harder for platforms to maintain their catalogs effectively.

In this project, we address this issue by predicting missing attributes from product text:

- **Attributes:** Structured labels such as hierarchical categories (L0–L4 or Supergroup → Group → Module) and brand.
- **Text/Description:** Unstructured input such as product titles (*“Enclume Angled Pot Hook, Set of 6”*) or receipt-style descriptions (*“1 car mat set”*).

2 Dataset

We use the **IndoML Datathon 2024 dataset** on attribute–value prediction for e-commerce products. It consists of two phases:

- **Phase 1 Dataset:**
 - *Input:* Product titles (with optional store/manufacturer details).
 - *Output:* Five levels of hierarchical categories (L0–L4) and brand.
 - *Example:* “Enclume Angled Pot Hook” → L0: Home & Kitchen → L1: Kitchen & Dining → ... → Brand: Enclume.
- **Phase 2 Dataset:**
 - *Input:* Receipt-style product descriptions, with retailer (anonymized) and price.
 - *Output:* Supergroup, group, module, and brand.

- *Example:* “1 car mat set” → Supergroup: Automotive, Group: Automotive detail unknown total, Brand: Receipt All.

In our project, we aim to **jointly utilize both datasets** to build models that can predict the complete set of attributes, including the hierarchical category levels (L0–L4) from Phase 1 and the higher-level groupings such as Supergroup from Phase 2. This combined setting helps in leveraging complementary information from both product titles and receipt-style descriptions.

3 Project Roadmap

To complete the project, we propose the following roadmap:

1. **Data Understanding and Preprocessing:** Explore the Phase 1 and Phase 2 datasets, clean and prepare the product text, and organize the data into input–output pairs.
2. **Initial NLP Models:** Build simple NLP models that use basic representations of text to predict the attributes. These will serve as baseline systems for comparison with more advanced methods.
3. **Advanced Models:** Train modern language models (e.g., BERT) that can better capture the meaning of short and noisy text, and use them to predict multiple attributes at once.
4. **Working with Both Datasets:** First train and test models separately on Phase 1 and Phase 2. Then, in this stage, attempt to combine both datasets in some way to see if using information from both helps improve predictions.
5. **Evaluation and Error Analysis:** Measure performance using accuracy and F1-score, compare simple and advanced methods, and analyze errors such as unseen brands or very short descriptions.
6. **Final Report and Presentation:** Summarize results, highlight lessons from using both datasets, and prepare the final submission.